MILITARY BULLETIN

U.S. BUILDING CODES AND STANDARDS

AN OVERVIEW

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Foreword

Codes and Standards are critical elements in the construction program of the United States. Considerable confusion exists within the Department of Defense and other Federal agencies regarding the interface among Federal construction criteria, local building codes and non-government standards.

To alleviate some of the confusion, this military bulletin provides background information on the use of codes and standards. It also provides information on the agencies that publish standards and the standards they publish.

All comments should be directed to:

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U. S. Building Codes and Standards

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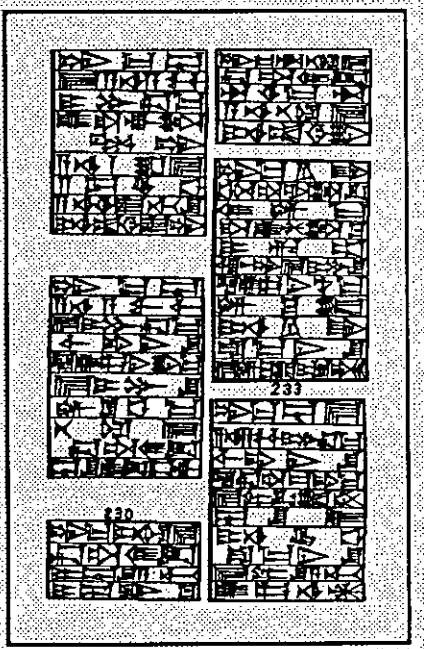
Codes and Standards Issuing Organizations B-1

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Building Codes Through History

The concept of building regulation is not new. In the Bible (Deuteronomy 22:8) there is a requirement for a house to have a parapet around the roof, so that the owner will not be judged liable if someone falls from his roof. Interestingly, the essence of this requirement continues in model building codes to this day. Building regulations have existed for millennia: they are found in the code of Hammurabi (1800 BC), in the Roman fire and sanitation codes of Augustus, Nero, and Vespasian (63 BC-79 AD) and the building standards known as Ying-tsao fa-shih published in 1100 AD during the Sung Dinasty in China.

Numerous rules for building construction from the Middle Ages to the present have addressed both structural safety and fire protection. Most faded from use as cities grew rapidly and the pressure for accommodations forced waivers. The first building regulations in America were established in New Amsterdam in 1625 while the first U.S. Federal Building regulations were adopted in 1791 for the District of Columbia. The first municipal building code in the United States was adopted by Chicago in 1875 as a response to the fire of 1871. The insurance industry became a strong advocate for building codes and in 1905 a recommended building code was issued by the National Board of Fire Underwriters. As interest in building regulation increased, municipal and state officials realized their efforts would be more efficient if they shared their experience. The first *model* code was published in 1927 by building officials in the western states.



King Hammurabi's Building Code

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— An Overview —

This bulletin describes the role of building codes and standards in planning, design, and construction. Whether the project is remodeling a bungalow, or constructing a major airport complex, all those involved—architects, engineers, contractors, artisans, suppliers, and ultimately the owner—are affected by building codes, and the standards which the codes reference. At some point in every project each player involved is likely to confront a technical problem. Often when technical problems are encountered, the player's perceive the applicable code or standard as the cause. More often, however, it is the code or standard that guides the players toward demonstrably effective solutions. Too often overlooked is the special value of codes and standards as the collected experience of the thousands of our peers.

Building codes and standards continually remind us of the high priority that must be assigned to safety. Aesthetics, productivity, marketability, and other factors may loom as major considerations, but all of them rank below safety as a priority for occupants and their neighbors.

"All sectors of the construction community benefit from building codes and related standards"

All sectors of the construction community benefit from building codes and related standards. They provide the basis of communication among the architect, contractor, suppliers, and building official. Even in the restricted situation where the building designer is associated with the constructor and the facility is not regulated, use of a building code will provide a basis for design validation and comparison with other facilities.

Conformance to certain standards can favorably influence insurance costs. When a building owner is confident that his newest facility meets applicable standards, communication with potential users and insurers is more reliable and authentic.

Model Building Codes

It is important to keep in mind the difference between a building code and a *model* building code. A building code is the legally adopted document in a state or other jurisdiction which governs construction—residential, commercial, industrial, and institutional—within its purview.

With few exceptions, all states, counties, and cities with building codes base them on one of the three well known model building codes. These are:

- National Building Code (BOCA)
- Standard Building Code (SBCC)
- Uniform Building Code (ICBO)

Complete addresses of the sponsoring organizations are given in Appendix A. These model codes represent a consensus of the building officials in a given region. The map in figure 1 illustrates regional affilia-

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tions with the model code bodies. Many states leave the option for affiliation to local jurisdictions, hence the 14 states shown as using "other combinations." The influence of a single model code in a state does not necessarily mean that there is a state-wide building code; it only means that all localities start with the same basis and consider the same amendments. These are modified by local jurisdictions to accommodate climatic or geographic conditions and local preferences. It is estimated that there are 44,000 Federal, state and local bodies which develop, change, and enforce regulations governing buildings.

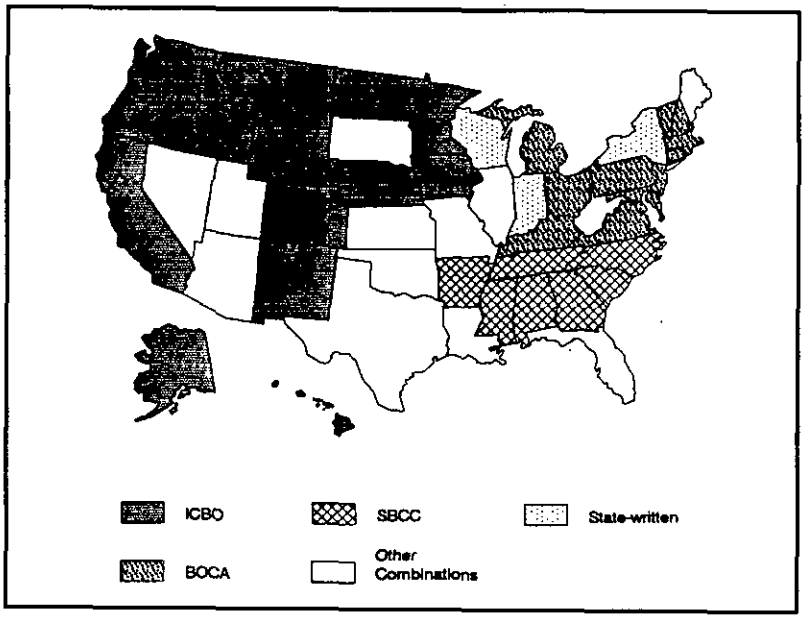


Figure 1. Regional Affiliations

"The NEC is the model on which virtually all state and local electrical codes are based."

Note that only New York and Wisconsin have state-written building codes. This means only that their officials do not participate directly in the model code deliberations and does not mean that they ignore the changes going on in building technology (see figures 2 through 4) . The cities of New York, Chicago, Los Angeles, and San Francisco are in similar positions. Each of the three regional model code bodies prepares other documents as supplements to the structural code.

Mechanical and plumbing codes are usually considered for adoption at the local level. Requirements of the mechanical code apply to heating, ventilating, and air conditioning; those of the plumbing code apply to water, gas, and sanitary service.

Other Model Code Bodies

Two major model code bodies operate nationally, neither of which are solely the province of building officials. They are the National Fire Protection Association (NFPA) and the American Society of Mechanical Engineers (ASME). NFPA is a technical society with broad membership, and it sponsors National Electrical Code (NEC) and the Life Safety Code (NFPA 101).

ASME performs the same functions for the Boiler and Pressure Vessel Code (BPVC).

The NEC is the model on which virtually all state and local electrical codes are based. A few major cities do not refer to the NEC in their adoption process, but the results closely parallel the NEC. This situation is true even when

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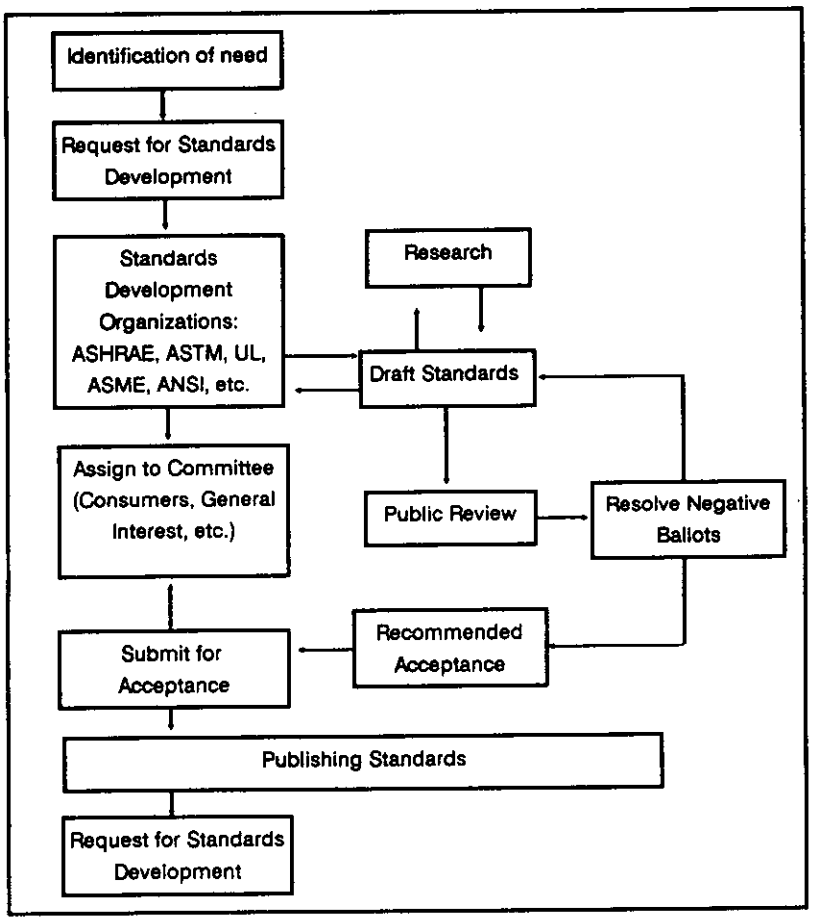


Figure 2. The Standards Development Process

the same city has a structural code which does not parallel a model building code.

Because of the close relationship between building electrical service and fire prevention, use of many parts of the NEC as product standards is extremely widespread. It is fortunate that the United States enjoys fully coordinated, nearly uniform regulation of electrical distribution and application as a result of the broad participation of all affected interests in development of the NEC.

The Life Safety Code (NFPA 101) is concerned with fire prevention as a means for minimizing personal injury. It was first developed at the time when building codes were primarily concerned with property protection.

Over the years some sections of NFPA 101 and the model building codes have come in conflict. The Life Safety Code is rarely adopted by local governments; however, the Health Care Financing Agency (HCFA) now requires that any health care facility must be designed to NFPA 101 requirements to qualify for Medicare/ Medicaid reimbursements. Since HCFA funds are distributed at the state level, many states have also adopted NFPA 101. The specific conflicts which occur between the state or local building code and NFPA 101 could be solved by requiring that facilities for institutional occupancies meet the tighter requirement.

The Boiler and Pressure Vessel Code (BPVC) is of interest to Federal agency designers when proposed construction includes steam or other industrial processing capability. The BPVC is typically required by insurance companies as a condition for underwriting equipment which is potentially explosive or implosive. Even when an organization self-insures, the use of a standard design guide is beneficial because of the built-in basis for comparing loss experience.

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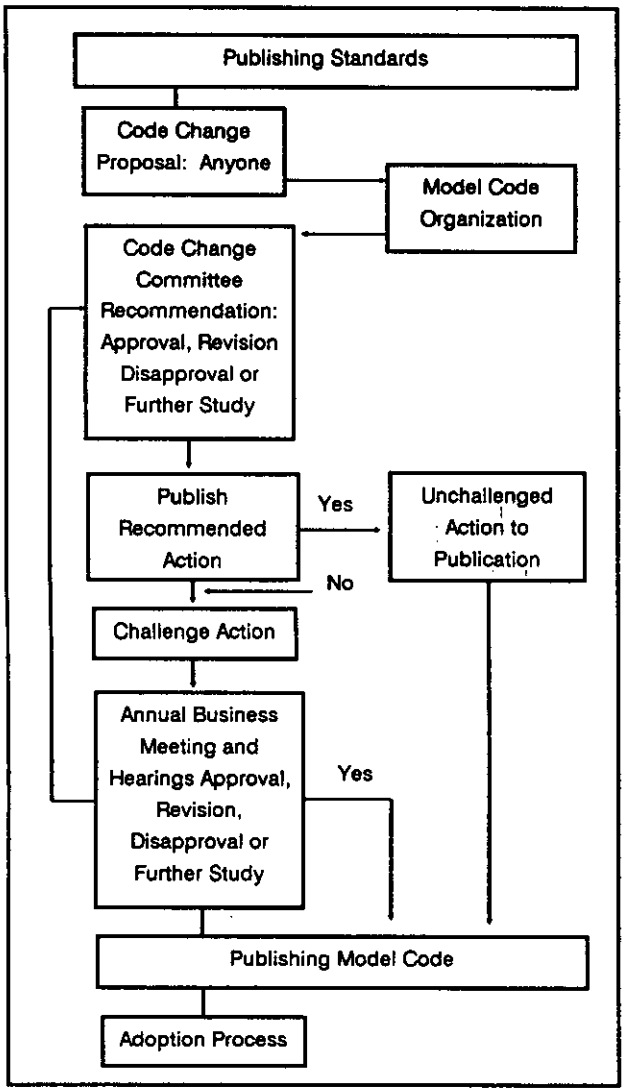


Figure 3. The Model Code Development Process

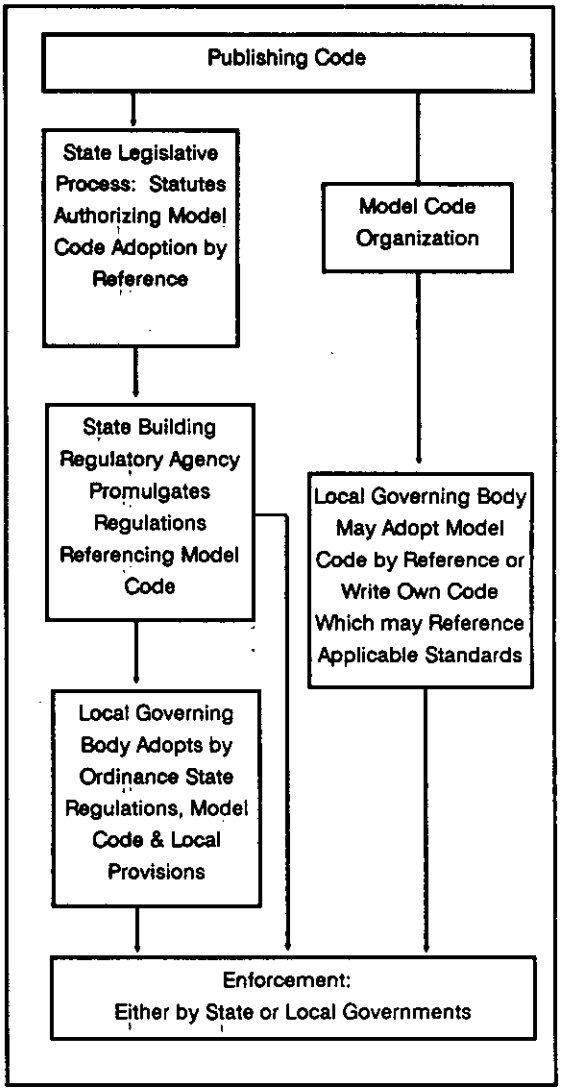


Figure 4. The Model Code Adoption Process

Appendix A gives some idea of the number of model codes in use. The presence of so many codes could lead to a certain amount of confusion among designers and suppliers, particularly those who operate on a national level. If suppliers had to manufacture to the variety of requirements and interpretations, this would lead to the possibility of many types of products to suit each code. Fortunately each code specifies product requirements by invoking appropriate standards. Each code lists those standards that are allowed for use, and these lists are generally the same in each code.

Government Standards

Nearly all Federal agencies develop and use standards concerning safe occupancy of buildings. On the civilian side,

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The Contentious World of Building Construction

Litigation is always a concern in the building industry. Usually conformance with acceptable industry practice is a tenable defense. Conformance to codes and generally accepted standards is excellent evidence of such acceptable industry practice. However, nonconformance does not always have to be an outright violation of the codes and standards, but can relate to interpretation or oversight.

For example, an interpretation of the barrier-free design code jeopardized the careers of a building official, architect and developer. This incident related to a four-story office building with basement parking. The first floor was several feet above grade so the architect, with the building official's agreement, placed the handicapped parking space in the basement with direct access to an elevator; a logical choice; but the code called for the main entry to be accessible to handicapped persons.

Shortly after the building opened, its owner, who had offices there, preempted the handicapped parking space and removed its signs. Soon thereafter a tenant was interviewing applicants for an accounting position and one of the applicants happened to be in a wheelchair. When he arrived there was no handicapped parking or ramp to negotiate the stairs to the main entry. Needless to say he did not make his interview or get the job. He did, however, hire an attorney and notify the city, building owner, and architect that he intended to sue. The attorneys for the persons who would have been sued advised them they might as well ask what he wanted and settle out of court.

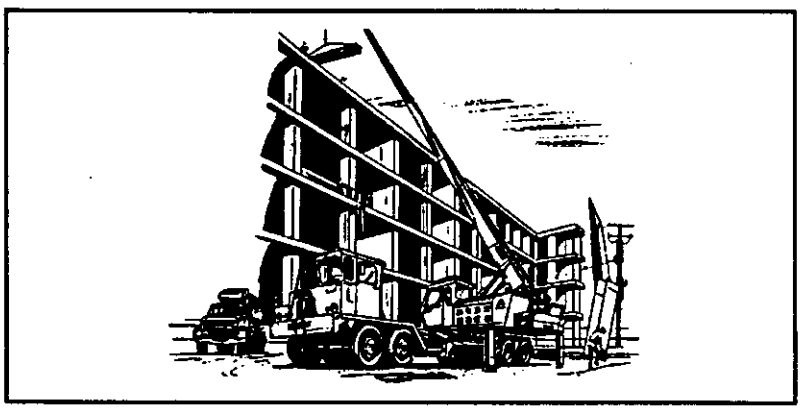
the General Services Administration (GSA) coordinates and promulgates all Federal specifications and standards. The distinction between these two kinds of documents is that the first mainly supports acquisition of products, while the second covers processes and practices that have been adopted as standard. Standards also establish design criteria, and requirements for selecting and applying products. If a building code were to exist for Federal agencies it would probably be a Federal Standard. Within it there would be references to Federal Specifications, many of which contain references to private voluntary standards.

On the military side, the Department of Defense (DOD) has established procedures for the development and coordination of military standards and specifications. Here the distinc-



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tion is the same as for civilian documents. Agencies within DOD provide the impetus for specific standards projects, while the urgency and scope dictate the degree of coordination required. The Army Corps of Engineers and Naval Facilities Engineering Command, being the primary construction organizations within DOD, sponsor those military specifications they need, but which are not available elsewhere. Both organizations have developed comprehensive sets of guidelines for their designers and contractors. These Guide Specifications and Technical Manuals provide proven solutions to design problems encountered in conventional construction as well as the specialized projects of the Armed Services. Like building codes, these guide specifications and technical manuals make extensive use of standards and specifications to define materials, products, test methods, and similar requirements. The more common building and construction requirements are defined by private sector standards, while military specifications and standards are used to document the more specialized requirements of the military.



In addition to GSA and DOD, thirty other government agencies and departments develop standards for their specialized building and construction needs. These include the Postal Service, Bureau of Indian Affairs, Occupational Safety and Health Administration (OSHA) and the U.S. Forest Service.

Private Sector Standards

The development of product standards in the private sector began in the engineering and technical societies such as ASME and the American Society for Testing and Materials (ASTM). As the use of building materials became a national, rather than regional phenomenon, the specialized trade associations began to supply documents to ensure common terminology and test procedures for use by both buyer and seller. Many of these documents have become national consensus standards through the procedures of the American National Standards Institute (ANSI).

Figure 5 indicates the relative number of standards that are available to the building and construction industry. Approximately 400 organizations developed the 35,000 private sector standards. Nearly half of those 400 organizations developed about 6,000 standards used in building and construction. The number of standards developed by these organizations range from ASTM's 620 (of its 7,500 standards) referenced in building codes to the three active standards of the Associated Locksmiths of America. Many organizations that develop standards have related interests. For example, listed below are fourteen developers of standards in the area of ventilation and air circulation. As might be expected, a good number of these standards are

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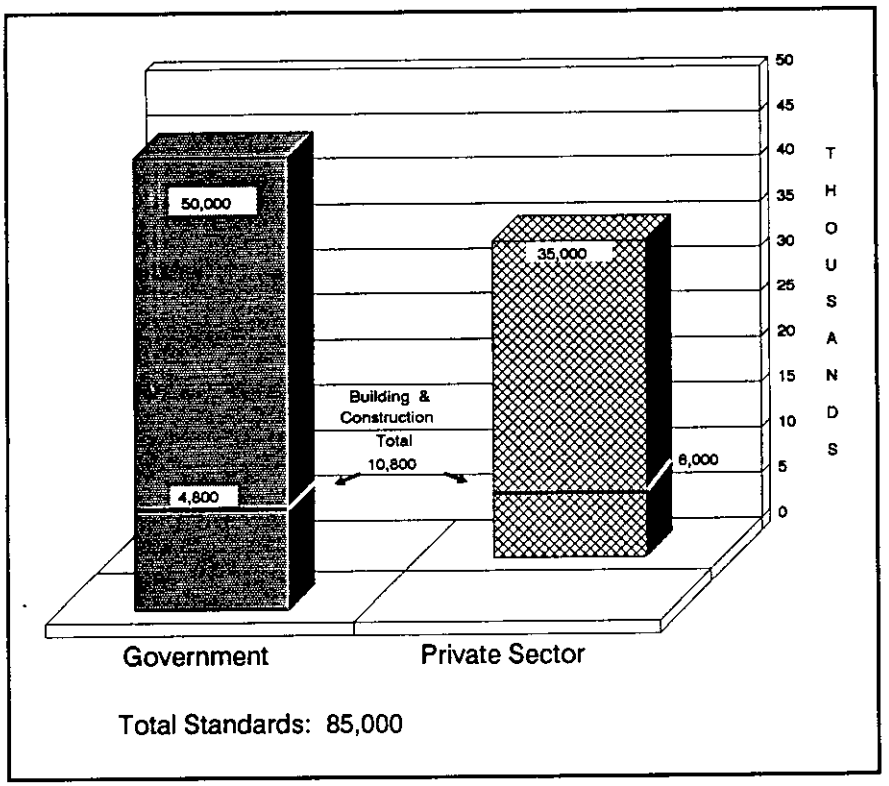


Figure 5A. U.S. Standards – All Types

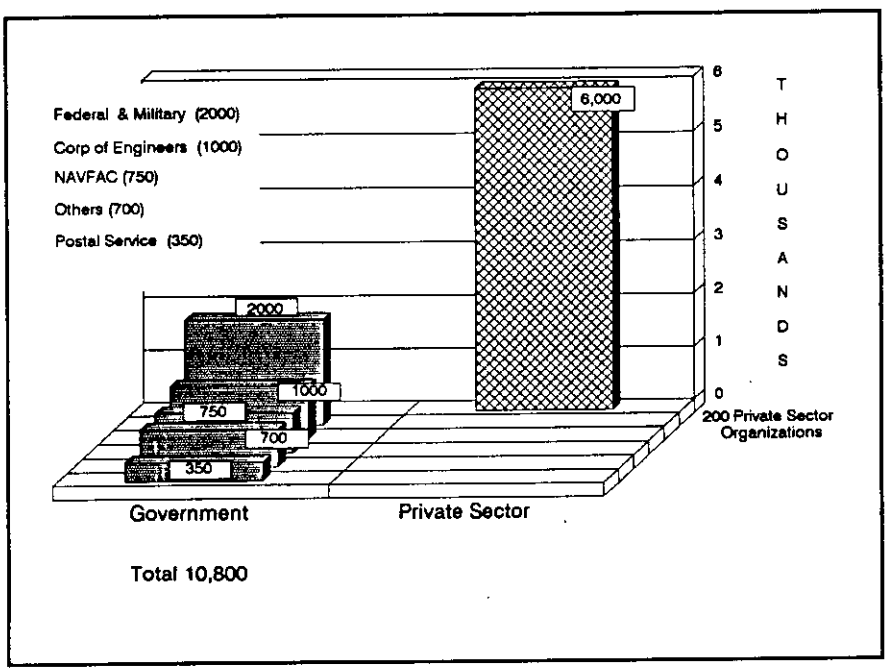


Figure 5B. U.S. Standards – Building and Construction

redundant or overlapping. Appendix B lists over 90 standards-producing organizations concerned with building standards for commercial building construction.

1. Air Diffusion Council
2. Air Distribution Institute
3. Air Movement & Control Association
4. American Fan Association
5. Associated Air Balance Council
6. Home Ventilating Institute
7. National Air Filtration Association
8. National Environmental Balancing Bureau
9. Underwriters Laboratories
10. Air-Conditioning Contractors of America
11. Air-Conditioning & Refrigeration Institute
12. American Society of Heating, Refrigerating & Air-Conditioning Engineers
13. Mechanical Contractors Association
14. Sheet Metal & Air-Conditioning Contractors National Association

Approval Agencies

Model building codes typically deal with product acceptance by delegating the responsibility to the "authority enforcing the code." (See figures 2 through 4

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Building Codes and Federal Agencies

Building codes in all of their forms are the result of local or state legislative action. The fundamental purpose of these technical requirements is to safeguard life, health, public welfare, and property. In general, any construction within the boundaries of the jurisdiction adopting a building code is subject to those rules. Frequently, property owned by the Federal government is exempt from the purview of the jurisdiction surrounding it. There is increasing awareness, however, that in many circumstances it is beneficial for Federal agencies to work with state and local governments in matters affecting public welfare, such as environmental protection and building regulation. This willingness to cooperate as good citizens is due to several factors: (1) Federal construction – civilian and military – is often adjacent to other private and public buildings; (2) Federal agencies lease space from private developers; and (3) State and regional agencies bear the responsibility for public welfare regulations that emanate from the Federal government.

It is now widely recognized in the Federal government that many installations conduct activities just like industrial facilities – be they offices, laboratories, warehouses, or processing facilities. It is becoming widely accepted that design and construction of such buildings should be performed in a manner consistent with civilian practice, even though plan review must be conducted within the particular Federal construction agency. Public trust and respect for the activities conducted at military installations is higher when the public perceives that the associated construction conforms to civilian requirements. Many military buildings are not solely occupied by military personnel. Such buildings are accessed by dependents, retirees, civilian contractors, and often the public on occasions such as "open house" tours. These kinds of occupancy must be considered in the initial design.

If the building is to be on existing Federally-owned land that has already been removed from local tax rolls, it is probable that no local or state building code will be mandated. The sponsoring agency, however, may have its own policy covering its role as a "good citizen" and thus voluntarily use the applicable building code. If the facility will be connected to local electrical, communications, gas, water, or sanitary facilities, applicable safety codes and related installation requirements may be specified by the utility providing the service.

Privately-owned property occupied by a Federal agency is *not* exempt from state or local building codes. In this age of litigation over issues of health and safety by civilian employees and even military personnel, building designers should resist any attempt to negotiate away any building code provisions required for property to be leased by their agency. In severe situations even occupants of nearby buildings, let alone other tenants, may object legally to the presence of an unsafe or nonconforming building.

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for typical codes.) This burden usually falls on the local government agency which has adopted the building code, be it structural, electrical, mechanical or plumbing. Fortunately the model code bodies provide an acceptance service that will receive test reports by independent laboratories from vendors and will review the data, then publish lists of products which conform to code requirements. Most local code administrators will approve products recognized by the model code body whose baseline document they adopted.

This technique can lead to regional differences, so the Council of American Building Officials (CABO) formed the National Research Board to provide a cross-recognition of building products recognized by the three model code bodies. NFPA does not participate in this service, however, preferring to delegate this function to other organizations. Until recently, Underwriters Laboratories, Inc. (UL) was the sole evaluation and listing service cited in most codes and standards. Now more and more states and cities accept the listing reports of several independent validation services.

The Department of Housing and Urban Development (HUD) determines the technical suitability of new products where no standards exist. This program results in Materials Releases, and Use of Materials Bulletins. HUD also provides for acceptance through independant validation organizations. Until a few years ago HUD maintained a comprehensive set of standards. Now it defers to local and model codes. The few HUD Minimum Property Standards only cover situations not addressed by existing codes and standards, or complement standards for health and safety with requirements to assure an adequate level of durability.

Coordinating Organizations

In recent years many organizations in the construction community have recognized the need to cooperate. The sheer size of the industry and the great number of participants dictate the need for formal coordination. Congress chartered the National Institute for Building Sciences (NIBS) in 1974. Now self-supporting, NIBS functions effectively to bring together Federal agencies, designers, developers, and trade associations to harmonize their responses to critical common issues. There is a steady stream of reports and seminars conducted by NIBS on major national construction issues.

Before the creation of NIBS, state officials concerned with construction policy began to meet, at the behest of the National Bureau of Standards, for the purpose of coordinating their actions. The meetings led to the formation of the National Conference of States on Building Codes and Standards (NCS/BCS). This organization provides a voice for the state officials in other forums, and also conducts some auxiliary programs for them.

Fourteen Federal departments and agencies responsible for the design, construction and operation of Federal government facilities address mutual problems and needs through the Federal Construction Council. Since 1953 the Council has addressed broad issues that affect the nation's largest builder, tenant, and landlord. Its fundamental purpose is to help Federal agencies having an interest in construction identify and find solutions to common problems. The Council promotes inter-

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agency cooperation in dealing with routine but troublesome tasks like the preparation of specifications and cost estimates and the introduction and use of computers in Federal construction agencies.

Among private sector standards developers, The American National Standards Institute (ANSI) serves as the umbrella organization. Nearly 50 trade associations, professional societies, and standards-developing committees coordinate their efforts through participating in ANSI's Construction Standards Board. A status report summarizes development and provides insight on the breadth of standards activities affecting buildings and construction.

Future Directions

Whether a neophyte or an "old hand," nearly everyone involved in building and construction recognizes that codes and standards have significant impact in all aspects of design, construction and operation. At the same time these observers also recognize that the codes and standards "system" is com-

Don't Codes and Standards Stifle Innovation and Increase Costs?

Modern building codes are amended regularly by the jurisdictions which use them. In this process the authorities are using the material developed by the technical organizations through which they collaborate on a regional basis. In this manner the building codes are as up-to-date as any other technical regulation.

In recent years collaborative efforts on building codes have become more performance-oriented, and less design-oriented. If particular portions of a building must be fire-resistant, material which meets a specified test requirement will probably be acceptable today. In past years fire-resistance requirements were met only by one or two well qualified materials. Some jurisdictions will be more conservative than others, particularly when it comes to evaluating the health effects of plastic materials subjected to heat for long periods of time. Such effects are not yet well understood and not included in any code.

The movement to procure items in a less prescriptive manner requires the buyer to be more informed about alternatives which might be available. Purchase specifications were often written to require identity with a preferred brand, or conformance with a standard, but allowed "any approved equivalent." This approach provided legal protection for the buyer and placed the burden of proof on the seller. Today if an innovator in building materials or a creative architect wants to introduce new techniques, he must be fully informed about all applicable test methods.

The fundamental purpose of building codes is to protect people and property. No responsible designer wants to abrogate these obligations. The basic concerns for structural adequacy, fire resistance, ventilation, and sanitation cannot be reduced to save costs. Building codes address these factors more clearly today than ever before and provide alternative solutions to traditional techniques and materials.

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plex; finding and interpreting applicable documents is difficult; and not only are there many players, but there are many games underway in the codes and standards arena.

In the 1930's, when life was a lot simpler than today, a number of organizations, particularly the National Bureau of Standards and ANSI (then called the American Standards Association), initiated actions that would result in a single national building code and a complementary, comprehensive set of national standards. The Depression and World War II slowed and sidetracked this effort. The post-war building boom created a need for more and better standards, and left little time for coordination and planning on a national level.

In recent years NIBS, NCS/BCS, the American Institute of Architects, and others have focused on ways to make the codes and standards more efficient, more effective, and more responsive to the needs of all segments of the building and construction community, including owners. The Federal Construction Council has supported these efforts and the Department of Defense is working actively within the system to improve the system.

A major initiative within DOD is to adopt non-government standards rather than develop or maintain military standards and specifications. This policy is detailed in Office of Management and Budget Circular A-119, and DOD Directive 4120.3. Building and construction has been a particularly fruitful area in which to apply this policy. Ultimately DOD, along with many others, would like a single national building code. While this may be a very difficult goal, it is not unattainable, but its accomplishment will certainly take time. In the interim, and possibly as a prerequisite to a single national code, is a need to streamline the process by which codes and standards are developed and implemented. DOD is working with the codes and standards community to improve the effectiveness of our codes and standards system because it makes good sense to realize the full potential from this national resource.

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APPENDIX A

Locating and Obtaining Relevant Codes, Standards, and Related Publications

A Comprehensive Database on Optical Disc

The standards search-aid for construction materials most useful to designers and contractors, particularly those involved in government projects, is the Construction Criteria Base, assembled and distributed by the National Institute of Building Sciences (NIBS). This database of more than 150,000 pages of information includes Navy Facilities Engineering Command, Corps of Engineers, DOD, NASA, and Veterans Administration construction specifications. It provides full text search and retrieval of any or all of these documents. The medium is a CD-ROM optical disc. The database is usable with IBM-PC XT or compatible computers. Once the desired information is located, it may be printed, copied to disks, or copied into a word processing file. A comprehensive specification processor called SPECSINTACT, is also available from NIBS.

Subscriptions to this database are annual with automatic quarterly updates. There are plans for increasing the amount of data applicable to economic analyses and including nongovernment codes and standards.

Hybrid Search and Storage Systems

All other search capabilities for any codes and standards, including those for construction materials and practices, are based on the document titles. One must locate the complete text separately either in microfilm or on paper in order to identify requirements or cite references. There are several commercial services for searching titles. These studies can be conducted through printed indexes, on-site computer files, or on-line computer files. One of the most popular of these services includes various combinations of microfilm with printed indexes.

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Another search aid which can help assure a designer that no possible source of specifications is being overlooked is the World Index of Building Certification Programs prepared by the Department of Housing and Urban Development. This document organizes standards development organizations into several dozen product categories including plant equipment as well as construction materials. It includes certification programs for standards and codes of both government agencies and private bodies in the United States and other leading industrial nations.

It is important to be familiar with the Department of Defense Index of Specifications and Standards (DODISS). This reference lists all DOD and GSA specifications, standards, and product descriptions, whether or not they are primarily intended for construction.

Codes

The model building codes are available from their respective publishers. Each design office should have a complete set of at least one of these series as well as a national electrical and plumbing code. The complete set will include such auxiliary documents as sign, swimming pool, and demolition codes.

The National Fire Code is the complete set of codes and standards prepared by the National Fire Protection Association. It is published as several soft-bound volumes. Many of the documents deal with management and operations of a local fire service and these practices may not be needed by design offices. Each document in the National Fire Code is available separately.

Standards

The primary source of standards is the sponsoring organization. All sell individual paper copies and bound volumes. There are at least two commercial organizations which maintain and distribute standards on microfilm. Typically their services are available on a subscription basis, and updates are often available sooner than from the hard-copy publisher.

A significant number of the rating guides and recommended practices for building materials are not broad consensus docu-

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ments. This information is generally not as readily available as the more well-known standards.

Specific Aid

The most widely-used specification cataloging system in the United States is that promulgated by the Construction Specification Institute (CSI). The major categories are recognized throughout the construction-supplies industry, and some vendor literature is printed with the CSI Identification on it.

This system is adaptable to decimal extensions by its users. Such a practice would be necessary if a design office kept a very large file of standards, guides, and catalogs in the CSI manner.

CSI, working with Computer Aided Design Information System, Inc. (CADIS) has developed a computerized information database with computerized graphics as well as text.

Approved Products Lists

The principle behind approving products for use in equipment and systems has the same source as building codes – the need for insurers to manage their risks. Today each of the model code bodies and the National Research Board examine test reports submitted by manufacturers and assemble lists which identify products that conform to code requirements. Such lists are not exhaustive, nor are they entirely exclusive.

Electrical equipment and some fire protection products are published by Underwriters' Laboratories (UL) as product lists. These are quite extensive and indicative of competing sources for a national market. Most other private certification programs utilize this extensive source of test results whenever appropriate.

Many of the code administrators in larger cities also maintain product recognition services. These lists often reflect local experience with equipment in those climates, sometimes being less inclusive than the regional or national lists.

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Qualified Products Lists (QPLs)

The Department of Defense and General Services Administration publish Qualified Products Lists (QPLs) of those items which comply with selected specifications. All specifications for products do not have QPLs. Less then five percent do. The primary criteria for establishing a QPL are the critical nature of the product, and the complexity and expense of performing comprehensive acceptance tests. Approximately 100 QPLs are maintained for building and construction products. They include fire resistant plywood, specialized floor surfaces, insulation for pipes, security hardware, and household lights and fluorescent fixtures. The largest single category is for paints and related protective finishes. All QPLs are listed in the DODISS.

On civilian projects contractors are obligated to use only products listed as approved by local jurisdictions. Similarly contractors on Federal projects may only use products listed on QPLs when the applicable specifications are cited in the design package. It is well known today that the building designer who wishes to use innovative techniques or materials may have to encourage vendors to submit samples for tests. Time must be allowed for this process if thorough documentation of materials and sources is to be accomplished.

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Organizations Issuing Building Codes and Standards

The publications listed below may be ordered directly from the issuing organizations which are listed in Appendix B.

	Issuing Organization	Document
Building Codes	BOCA	National Building Code
		National Existing Structure Code
	ICBO	Uniform Building Code
	SBCC	Standard Building Code
	CABO[1]	One & Two Family Dwelling Code
	NFPA	Life Safety Code, 101
Plumbing Codes	BOCA	National Plumbing Code
	IAPMO[2]	Uniform Plumbing Code
	SBCC	Standard Plumbing Code
	NAPHCC	*National Standard Plumbing Code*
Mechanical Codes	ASME	Boiler and Pressure Vessel Code
	BOCA	National Mechanical Code
	ICBO/IAPMO[3]	Uniform Mechanical Code
	SBCC	Standard Mechanical Code
Fire Codes	BOCA	National Fire Prevention Code
	ICBO/WFCA	Uniform Fire Code
		Uniform Fire Code Standards
	SBCC	Standard Fire Prevention Code
	NFPA	National Fire Prevention Codes (260 codes and standards, plus recommended practices and guidelines)

[1]Also available from BOCA, ICBO, or SBCC

[2]Endorsed by ICBO

[3]Joint publications

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	Issuing Organization	Document
Housing Codes	BOCA	Basic Property Management Code
	ICBO	Uniform Housing Code
	SBCC	Standard Housing Code
Energy Conservation Codes	BOCA	National Energy Code
	CABO	Model Energy Code (based on ANSI/ASHRAE/IES 90)
	IAPMO	Uniform Solar Energy Code
Electrical Codes	ANSI	National Electrical Safety Code, ANSI C2
	ANSI/NFPA	National Electrical Code, 70
		One & Two Family Dwellings Electrical Code, 70A
Flammable Gases and Liquids Codes	SBCC	Standard Gas Code
	NFPA	Automotive and Marine Service Station Code, 30A
		Flammable and Combustible Liquids Code, 30
		National Fuel Gas Code, 54
Swimming Pool Codes	SBCC	Standard Swimming Pool Code
	IAPMO	Uniform Swimming Pool, Spa & Hot Tub Code
Miscellaneous Codes	ANSI/ASME	Safety Code for Elevators and Escalators, A 17.3
	ANSI/NFPA	Lightning Protection Code, 78
	BOCA	National Private Sewage Disposal Code
	ICBO	Dwelling Construction Under the Uniform Building Code
	ICBO	Uniform Building Security Code
	ICBO	Uniform Code for the Abatement of Dangerous Buildings
	ICBO	Uniform Sign Code
	SBCC	Standard Unsafe Building Abatement Code
Building Code Interpretation Manuals	BOCA	Basic Building Code Interpretations
	ICBO	Uniform Building Code Application/Interpretations Manual
	SBCC	Standards Code Interpretation Manual
	IAPMO	Uniform Plumbing Code Study Guide
	NFPA	Life Safety Code Handbook
	NFPA	National Electrical Code Handbook
	CABO	Application & Commentary – One & Two Family Dwelling Code

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	Issuing Organization	Document
Federal Design Criteria and Standards	COE	U.S. Army Corps of Engineers Construction Guide Specifications
		U.S. Army Corps of Engineers Technical Manuals
	FAA	Federal Aviation Administration Standards for Specifying Construction of Airports, AC/150/5370-10
	GSA	General Services Administration Specifications
		General Services Administration Design Manuals
	HUD	Housing and Urban Development Minimum Property Standards
		Housing and Urban Development Manufactured Home Standards
	NASA	National Aeronautics And Space Administration Master Construction Specifications
	NAVFAC	Naval Facilities Engineering Command Guide Specifications
		Naval Facilities Engineering Command Design Manuals
	USPS	Standard Design Criteria
	VA	Veterans Administration Master Specifications
Building Standards	These publications represent some of the broader standards having wide application.	
	ANSI	Method of Measuring Floor Space in Office Buildings, Z65.1
		Minimum Design Loads for Buildings and Other Structures, A58.1
		Providing Accessibility and Usability for Physically Handicapped People, A117.1
	ANSI/ASHRAE/IES	Energy Conservation in New Building Designs, 90A
		Energy Conservation in Existing Buildings, High Rise Residential, 100.2
		Energy Conservation in Existing Buildings, Institutional, 100.5
		Energy Conservation in Existing Buildings, Public Assembly, 100.6
	ANSI/ASME	Safety Code for Elevators and Escalators, A17.1
	ANSI/IEEE	Recommended Practice for the Design of Reliable Industrial and Commercial Power Systems, Std 493

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	ISSUING ORGANIZATION	DOCUMENT
Building Standards		Recommended Practice for Electric Power Distribution for Industrial Plants, Std 141
		Recommended Practice for Electric Power Systems in Commercial Buildings, Std 241
		Recommended Practice for Electric Systems in Health Care Facilities, Std 602
		Recommended Practice for Energy Conservation and Cost-Effective Planning in Industrial Facilities, Std 739
		Recommended Practice for Emergency and Standby Power for Industrial and Commercial Applications, Std 446
		Recommended Practice for Grounding of Industrial and Commercial Power Systems, Std 142
		Recommended Practice for Industrial and Commercial Power System Analysis, Std 399
		Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems, Std 242
	ANSI/NFPA	Health Care Facilities, 99
		Parking Structures, 88A
		Storage and Handling of Liquefied Petroleum Gases, 58
	ASTM	Standards in Building Codes, 620 standards called out in BOCA, ICBO, SBCC, and CABO codes and the National Code of Canada, three volumes, PCN: 03-002588-10
	CABO	Recommended Requirements for Code Officials for Solar Heating, Cooling and Hot Water Systems
	ICBO	Uniform Building Code Standards
	NFPA	Emergency and Standby Power Systems, 110
	SBCC	Standards for Use with Standard Building Code
		Standard for Installation of Roof Coverings
		Standard for Flood Plain Management
		Standard for Walls in Hurricane Force Winds
	UL	Building Materials, Q116
		Fire Resistance, Q118

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List of Acronyms

AIA	American Institute of Architects
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigerating & Air-Conditioning Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
BOCA	Building Official & Code Administrators International
CABO	Council of American Building Officials
COE	Corps of Engineers
FAA	Federal Aviation Administration
FCC	Federal Construction Council, National Academy of Sciences
GPO	Government Printing Office
GSA	General Services Administration Public Buildings Service
IHS	Information Handling Services
HUD	Department of Housing & Urban Development
IAPMO	International Association of Plumbing and Mechanical Officials
ICBO	International Conference of Building Officials
IEEE	Institute of Electrical and Electronics Engineers

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IES	Illuminating Engineering Society
NASA	National Aeronautics and Space Administration
NAPHCC	National Association of Plumbing - Heating - Cooling Contractors
NAVFAC	Naval Facilities Engineering Command
NCS\BCS	National Conference of States on Building Codes and Standards
NIST	National Institute of Standards and Technology (Formerly National Bureau of Standards [NBS])
NFPA	National Fire Protection Association
NIBS	National Institute of Building Sciences
NSA	National Standards Association
SBCC	Southern Building Code Congress International Inc.
UL	Underwriters Laboratories
USPS	U.S. Postal Service
VA	Veterans Administration
WFCA	Western Fire Chiefs Association

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APPENDIX B

Standards Organizations and Other Document Sources

Depending on the definition used, approximately 200 organizations, mainly trade and professional associations and societies, produce building standards in the United States. Most of these organizations operate with considerable autonomy. This Appendix is concerned with only the most important building standards for commercial building construction and lists over 90 standards-producing organizations.

Unfortunately, there is no single source for information or copies of building standards. Major third party suppliers are listed at the end of this appendix.

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List of Codes and Standards Developing Organizations

Air Conditioning Contractors of America (ACCA)
1513 16th Street, NW
Washington, DC 20036
(202) 483-9370

Air-Conditioning and Refrigeration Institute (ARI)
1501 Wilson Boulevard
Arlington, VA 22209
(703) 524-8800

ARI is a trade association of manufacturers of HVAC equipment and an active standards developer. ARI standards are primarily directed toward evaluation of equipment performance. ARI frequently references the safety standards issued by other organizations. It serves as secretariat for several ANSI standards. ARI standards and other technical documents are distributed directly, and through ANSI, IHS, NSA, and Global.

Air Movement and Control Association
30 West University Drive
Arlington Heights, IL 60004
(312) 394-0150

Performs product certification for residential, commercial, and industrial fans, ventilators, air curtains, dampers, and shutters.

Aluminum Association (AA)
900 19th Street, NW
Suite 300
Washington, DC 20006
(202) 862-5100

AA represents virtually all finished-aluminum producers and about 85 percent of the semi-finished-aluminum producers. The Association produces a small number of standards which are processed through ANSI. Activities of the association form the basis for most aluminum standards of government agencies, ASTM, and company standards. Standards cover topics such as extruded and tubular aluminum products, a designation system for aluminum finishes, specifications for aluminum structures, and standards for sand and permanent mold aluminum castings. Also publishes Aluminum Standards and Data biennially.

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American Association of State Highway and Transportation Officials (AASHTO)
444 North Capitol Street, NW
Suite 225
Washington, DC 20001
(202) 624-5800

A state government association and standards organization that publishes the Standard Specification of Transportation Materials and Methods of Sampling and Testing. It also publishes standards for bridges and highways, and a policy on urban highways and arterial streets.

American Boiler Manufacturers Association (ABMA)
950 North Glebe Road
Arlington, VA 22203
(703) 522-7350

ABMA is the trade association for boiler manufacturers and allied products. Publishes fire boiler ratings and serves as the secretariat for NFPA Committee on Boiler Furnace Explosions.

American Concrete Institute (ACI)
22400 West Seven Mile Road
Detroit, MI 482199
(313) 532-2600

ACI standards focus on structures as a whole. Standards on basic materials and standard components are deliberately excluded. More than 30 ACI standards have been developed and all are routinely processed through ANSI. Twelve ACI standards have been adopted by the Department of Defense.

American Conference of Governmental Industrial Hygienists (ACGIH)
6500 Glenway Avenue, Building D-7
Cincinnati, OH 45211
(513) 661-7881

National professional association of health and safety officials. Publishes the Threshold Limit Values that form the basis for most regulations on industrial contaminants and the Industrial Ventilation Manual of Recommended Practices. Standards are regularly updated and sold directly.

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American Gas Association (AGA)
1515 Wilson Boulevard
Arlington, VA 22209
(703) 841-8400

A national association of gas transmission and distribution companies. Serves as secretariat for several ANSI committees for gas appliances (ANSI Z21), industrial gas equipment (ANSI Z83), and the National Fuel Gas Code (ANSI B109). Standards are distributed directly and through ANSI.

American Hardboard Association (AHA)
520 North Hicks Road
Palatine, IL 60067
(312) 934-8800

Represents hardboard producers, and develops standards for these products in their interior and exterior uses. The Association conducts a certification program. American Hardboard Association standards are developed and sold through ANSI.

American Industrial Hygiene Association (AIHA)
475 Wolf Ledges Parkway
Akron, OH 44311
(216) 762-7294

Promotes the study and control of environmental stresses arising in or from the workplace.

American Institute of Architects (AIA)
Building/Performance and Regulations Committee
1735 New York Avenue, NW
Washington, DC 20006
(202) 626-7448

American Institute of Steel Construction (AISC)
400 North Michigan Avenue
Chicago, IL 60611
(312) 670-2400

Trade association that produces the Specification for Structural Steel for Buildings and the Manual of Steel Construction. AISC standards are regularly updated. They are sold directly and through IHS.

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American Institute of Timber Construction (AITC)
11818 SE Millplain
Suite 415
Vancouver, WA 98684
(206) 254-9132

Concerned primarily with the development of standards for timber construction and laminated timber construction. One of its standards has been processed through ANSI (A190.1). AITC documents are sold directly.

***American Insurance Association* (AIA)**
85 John Street
New York, NY 10038
(212) 669-0400

Trade association for the insurance industry. Publishes the Fire Prevention Code, and many industry bulletins. Its documents are sold directly. The Association was the original publisher of the National Building Code, the first of the U.S. model building codes.

American Iron and Steel Institute (AISI)
1133 15th Street, NW
Suite 300
Washington, DC 20005
(202) 452-7100

AISI is a trade association producing three design specifications for light gage, and cold formed steel. Also develops standards for steel cable and fireproofing of steel members. Serves as secretariat for U.S. Technical Advisory Group for ISO Technical Committee 167 on Steel and Aluminum Structures. AISI standards are sold directly and through Global.

American Lumber Standards Committee (ALSC)
19715 Waters Road,
P.O. Box 210
Germantown, MD 20874
(301) 671-8680

This is a NIST Product Standards Committee that sets the standards under which most of the lumber in the United States is produced. It acts as the standing committee for revision of NIST Voluntary Product Standard 20-70, which classifies American Standard lumber. Licenses lumber grading agencies.

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American National Standards Institute (ANSI)
1430 Broadway
New York, NY 10018
(212) 354-3300

ANSI is mainly a standards coordinating and approval organization, and is the U.S. representative to the International Organization for Standardization. Over 8,500 standards are approved through ANSI procedures and are designated as American National Standards. Frequently standards have ANSI designations as well as designations from the standards organization that originally developed the standard and transmitted it to ANSI for national consensus review. In the case of most ANSI standards, another standards-writing organization serves as secretariat for the committee responsible for development and maintenance of the relevant ANSI standards. ANSI standards referenced in the Model Codes range from basic material standards such as those for tile, cement and adhesives, to whole-building standards for access for the physically handicapped, and building code requirements for reinforced masonry. All ANSI-approved standards are available through ANSI, through IHS, and are usually also available through the organization that originally developed the standard. The standards development process contains procedures for consensus review and the reconciliation of negative ballots. All ANSI standards are reviewed periodically and are either revised, reaffirmed, or discontinued.

American Petroleum Institute (API)
1220 L Street, NW
Washington, DC 20005
(202) 682-8000

API standards are used extensively in the petroleum industry rather than in building regulation.

American Plywood Association (APA)
P.O. Box 22700
Tacoma, WA 98411
(206) 565-6600

APA is an association of plywood producers. It produces standards for plywood and plywood uses, including design of plywood beams, stressed skin plywood, and plywood design specifications. APA standards are regularly referenced in building regulations and followed by many government agencies. These standards are sold directly. APA also serves as secretariat for the NIST Industry Standards Committee to keep PS 1/ ANSI A199.1 current.

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American Public Health Association, Inc. (APHA)
1015 15th Street, NW
Washington, DC 20005
(202) 789-5600

In the past APHA has produced the Standards for Healthful Housing, but this model document is now considered to be out of date and in need of revision. APHA standards are primarily concerned directly with health, with the most direct tie-in to building design and construction being methods for measuring contaminants in the air. APHA standards are distributed directly.

American Railway Engineering Association (AREA)
50 F Street, NW
Suite 7702
Washington, DC 20001
(202) 639-2190

Publishes information relating to the construction and maintenance of railroad fixed structures and rights of way.

***American Society for Testing and Materials* (ASTM)**
1916 Race Street
Philadelphia, PA 19103
(215) 299-5400

The Department of Defense references roughly 2,000 ASTM standards, while local building codes reference approximately 600. Because of the extensive adoption of ASTM standards within building regulations, ASTM periodically publishes a volume of ASTM standards referenced in building regulations. Most ASTM standards applicable to buildings are for basic material and component properties. Of special note are ASTM fire test standards for materials and assemblies. ASTM standards are developed through extensive procedures for consensus, and are updated periodically. ASTM standards are sold directly and through IHS and Global.

American Society of Civil Engineers (ASCE)
345 East 47th Street
New York, NY 10017
(212) 705-7496

ASCE is involved in many standards areas, but of special importance for building construction are ASCE developments relating to foundations, deck slabs, masonry construction, and structural assessment of existing buildings. Standards available through ASCE.

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American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
(404) 636-8400

ASHRAE is an international membership organization for HVAC systems. It produces many standards, technical reports and manuals for the design, testing and evaluation of these systems, ASHRAE standards for energy conservation in new and existing buildings and for ventilation are widely referenced in building regulations. The ASHRAE Handbook series is also referenced extensively in building regulations. ASHRAE standards and technical documents are updated periodically and are distributed directly through ANSI and through IHS and Global.

American Society of Mechanical Engineers (ASME)
22 Law Drive
Box 2300
Fairfield, NJ 07007-2300
(201) 882-1167
(800) 843-2763

ASME has over 700 active standards, of which over 80 have been adopted by the Department of Defense. The ASME Boiler Code is widely referenced in building regulations. ASME standards are routinely processed through ANSI procedures. They are available directly and through ANSI, IHS and Global.

American Society of Plumbing Engineers (ASPE)
3617 Thousand Oaks Boulevard, #210
Westlake, CA 91362
(805) 495-7120

ASPE is a technical society that actively participates in the standards activities of other organizations, especially the National Association of Plumbing-Heating-Cooling Contractors (NAPHCC), ANSI, IAPMO, and ASSE.

American Society of Sanitary Engineering (ASSE)
P.O. Box 40362
Bay Village, OH 44140
(216) 835-3040

ASSE is primarily concerned with the development of standards for plumbing system components. Products meeting these standards are identified by an ASSE seal. ASSE standards are widely referenced by government agencies. They are distributed directly as well as through IHS.

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American Water Works Association, Inc. (AWWA)
6666 West Quincy Avenue
Denver, CO 80235
(303) 794-7711

Standards relate primarily to pipe, fittings, valves, water treatment, water storage tanks and similar topics. Standards are distributed directly and through NSA, IHS, Global, and Dialog Information Services.

American Welding Society, Inc. (AWS)
550 N.W. LeJeune Road
P.O. Box 351040
Miami, FL 33135
(305) 443-9353

AWS produces welding standards, often in conjunction with ANSI and SAE. The standards are reviewed regularly and are sold directly and through ANSI, IHS, and Global.

American Wood Preservers Association (AWPA)
P.O. Box 849
Stevensville, MD 21666
(301) 643-4163

American Wood Preservers Bureau (AWPB)
P.O. Box 5283,
Springfield, VA 22150
(703) 339-6660

Standards produced by the American Wood Preservers Bureau focus on those for wood and plywood preservatives and treatment. AWPB runs a certification program with a quality mark that is widely recognized by building regulatory organizations and government agencies such as HUD, VA, and Farmers Home Administration.

American Wood Preservers Institute (AWPI)
1945 Old Gallows Road, Suite 550
Vienna, VA 22182
(703) 893-4005

One of the referenced standards for AWPI is for pole building design.

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American Architectural Manufacturers Association (AAMA)
2700 River Road, Suite 118
Des Plaines, IL 60018
(312) 699-7310

AAMA standards are principally concerned with windows, doors, and coatings and finishes for exterior aluminum. They are referenced by HUD in the MPS and are accepted by the Department of Defense.

Associated Air Balance Council (AABC)
1518 K Street, NW
Suite 503
Washington, DC 20005
(202) 737-0202

Independent testers of air-handling systems.

Associated General Contractors of America (AGCA)
1957 E Street, N.W.
Washington, DC 20006
(202) 393-2040

Membership of heavy-construction contractors.

Brick Institute of America (BIA)
(Formerly the Structural Clay Products Institute)
11490 Commerce Park Drive, Suite 300
Reston, VA 22091
(703) 620-0010

The Brick Institute is a national association of brick manufacturers involved in all significant standards related to the use of brick and masonry in construction. Brick Institute standards include those for engineered brick masonry and for sound insulation of clay masonry walls. The Institute maintains SCPI standards.

Building Officials & Code Administrators, International (BOCA)
4051 West Flossmoor Road
Country Club Hills, IL 60478
(312) 799-2300

Building Officials and Code Administrators International (BOCA) is one of the three Model Building Code organizations in the U.S.. It is a membership organization of building officials from jurisdic-

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tions principally in the eastern and midwestern states. BOCA published a full set of model building regulations including a building code, mechanical code, plumbing code, existing structures code and fire prevention code. Recently, BOCA has taken over responsibility for the National Building Code when the American Insurance Association decided to stop updating that document. Heretofore named the Basic codes, the BOCA codes are now termed the Basic/National Codes, and in the future they will be called the National series of codes.

Cast Iron Soil Pipe Institute (CISPI)
5959 Shallowford Road
Suite 419
Chattanooga, TN 37421
(615) 892-0137

Develops cast iron soil pipe standards that are referenced in building regulations and adopted by many government agencies.

Cedar Shake and Shingle Bureau
515 116th Avenue, NE
Bellevue, WA 98004
(206) 453-1323

Develops specifications for wood shakes and shingles.

Concrete Reinforcing Steel Institute (CRSI)
933 N. Plum Grove Road
Schaumburg, IL 60173
(312) 517-1200

An association of manufacturers of steel reinforcing bars, CRSI produces standards for reinforced concrete, including reinforced concrete fire resistance. Standards are sold directly.

Consumer Product Safety Commission (CPSC)
Office of the Secretary
5401 Westbard Avenue
Bethesda, MD 20207
(301) 492-6800

The Consumer Product Safety Commission is an independent Federal agency charged with protecting the public from unreasonable risk of injury from consumer products. CPSC develops standards only if it determines that existing standards do not provide adequate protection for the public. Upon adoption, CPSC standards are mandatory and preempt all other regulations of

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federal, state and local government that are less stringent. In building design, the most important CPSC standards are for architectural glazing and cellulose insulation.

Cooling Tower Institute (CTI)
616 FM1960 West
Suite 204
Houston, TX 77090
(713) 583-4087

Promotes improvement in technology, design, performance, and maintenance of cooling towers. Standards are available directly and through IHS.

Corps of Engineers Depot (COE)
2803 52nd Avenue
Hyattsville, MD 20781
(301) 436-2063

Council of American Building Officials (CABO)
5203 Leesburg Pike
Suite 708
Falls Church, VA 22041
(703) 931-4533

The members of CABO are the three model building code organizations. Through CABO they seek coordination in the development of their respective codes, and in cooperation with other organizations and the Federal government. CABO itself has been used as a vehicle for the development of the One and Two Family Dwelling Code and the Model Energy Code which in turn is based on the ASHRAE standard for energy conservation in new building design.

Factory Mutual Engineering Corporation (FM)
Standards Laboratories Department
1151 Boston Providence Turnpike
Norwood, MA 02062
(617) 762-4300

Factory Mutual is a leader in the area of fire testing, and the development of guidelines for fire loss prevention. FM participates in the standards development work of other organizations and prepares Approval Standards for assemblies such as steel decks.

Federal Aviation Administration (FAA) – Standards for airports and related structures are available from Superintendent of Documents, Government Printing Office.

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Federal Construction Council (FCC)
National Academy of Sciences
2101 Constitution Avenue, NW
Washington, DC 20418
(202) 334-2842

General Services Administration/Public Buildings Service (GSA/PBS)
7th and D SW
Room 2609
Washington, DC 20407
(202) 472-1490

The Federal specifications program was developed to provide coordination between military and civilian government procurement standards. In this process, over 300 voluntary private sector standards have been adopted for Federal government use.

Government Printing Office (GPO)
Superintendent of Documents
Washington, DC 20402
(201) 783-3238

Gypsum Association (GA)
810 1st Street, NE
#510
Washington, DC 20002
(202) 289-5440

Promotes the use of products containing gypsum. Develops and maintains standards for gypsum board finishing and application, and a fire resistance design manual. Gypsum Association standards are distributed directly.

Hardwood Plywood Manufacturers Association (HPMA)
P.O. Box 2789
Reston, VA 22090
(703) 435-2900

The Hardwood Plywood Manufacturers Association has developed standards relating to hardwood and decorative plywood, a structural design guide, and most recently has cooperated in the development of standards for testing the formaldehyde emissions of plywood. This test standard has been referenced in the HUD regulations on formaldehyde emission for manufactured housing.

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Housing & Urban Development, Department of (HUD)
451 7th Street, SW
Room B258
Washington, DC 20410-5400
(202) 755-6590

Hydronics Institute (HI)
(Formerly Institute of Boiler and Radiator (IBR) Manufacturers)
35 Russo Place
Berkeley Heights, NJ 07922
(201) 464-8200

This trade association for the heating industry develops standards for rating baseboard heating units. Steel Boiler Institute ratings SBI now included under IBR.

Illuminating Engineer Society (IES)
345 East 47th Street
New York, NY 10017
(212) 705-7920

Maintains approximately 80 technical committees and an active research program. Several IES standards have been processed through ANSI procedures, including those for office, school, industrial, and roadway lighting. IES standards are distributed directly and through ANSI, IHS and Global.

Institute of Electrical and Electronics Engineers (IEEE)
345 E. 47th Street
New York, NY 10017
(212) 705-7960

IEEE is a professional society, and a major standards producer for the electrical and electronics field. Some 30,000 IEEE members are active in standards development. The Institute produces approximately 500 standards. IEEE is accredited by ANSI, and many IEEE standards are processed through ANSI. IEEE serves as secretariat for approximately 17 ANSI standards. The National Electrical Safety Code, ANSI C2, is widely adopted by government and regulatory agencies. Several other IEEE standards have been adopted by DOD. IEEE standards are sold directly and through services such as Global and IHS.

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International Association of Plumbing and Mechanical Officials (IAPMO)
5032 Alhambra Avenue
Los Angeles, CA 90032
(213) 223-1471

International Conference of Building Officials (ICBO)
5269 South Workman Mill Road
Whittier, CA 90601
(213) 699-0541

ICBO is one of the three model building code organizations in the U.S. ICBO produces the Uniform series of building codes and companion codes. These model codes are updated periodically, and are approved based on voting by ICBO members.

International Masonry Institute (IMI)
823 15th Street, NW
Washington, DC 20005
(202) 783-3908

IMI has developed recommended practices and guide specifications for cold weather masonry.

Manufacturers Standardization Society of the Valve and Fitting Industry, Inc. (MSS)
127 Park Street, NE
Vienna, VA 22180
(703) 281-6613

The BOCA Mechanical Code references the MSS standard for pipe hangers and supports.

Maple Flooring Manufacturers Association (MFMA)
60 Revere Drive
Suite 500
Northbrook, IL 60062
(312) 480-9080

Establishes product groups and grades. Standards are distributed directly.

Mechanical Contractors Association of America (MCAA)
5410 Grosvenor Lane
Suite 120
Bethesda, MD 20814
(301) 897-0770

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National Aeronautics and Space Administration (NASA)
Kennedy Space Center
Code DE-C-1
Kennedy Space Center, FL 32899

Metal Building Manufacturers Association (MBMA)
1230 Keith Building
Cleveland, OH 44115
(216) 241-7333

National Association of Plumbing-Heating-Cooling Contractors (NAPHCC)
180 S. Washington Street
Box 6808
Falls Church, VA 22046
(703) 237-8100

The Association is a trade association that participates actively in the technical work of other standards writing organizations. It is also responsible for production of the National Standard Plumbing Code which was adopted by HUD in 1973. (Note, however, that the HUD Minimum Property Standards will not prevail when localities have adopted recent versions of one of the model building codes.) The National Standard Plumbing Code is maintained by a technical committee of the Association.

National Board of Boiler & Pressure Vessel Inspectors (NBBPVI)
1055 Crupper Avenue
Columbus, OH 43229
(614) 888-8320

National Bureau of Standards—See National Institute of Standards and Technology (NIST).

National Concrete Masonry Association (NCMA)
2302 Horse Pen Road
P.O. Box 781
Herndon, VA 22070
(703) 435-4900

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National Electrical Manufacturers Association (NEMA)
2101 L Street, NW
Suite 300
Washington, DC 20037
(202) 457-8400

National Fire Protection Association (NFPA)
Batterymarch Park
Quincy, MA 02269
(617) 770-3000
or (800) 344-3555

The National Fire Protection Association is a leading source in the U.S. for fire related standards. The NFPA Fire Codes are extensively referenced in building regulations as is the NFPA Life Safety Code. The full set of NFPA Fire Codes includes over 250 fire prevention, protection and suppression codes, standards and similar documents. Another product of NFPA is the National Electrical Code which is adopted almost universally across the United States. NFPA standards are regularly updated by approximately 175 technical committees. They are sold directly, and through IHS and Global.

National Forest Products Association (NFPA)
1250 Connecticut Avenue, NW
Suite 200
Washington, DC 20036
(202) 463-2700

The National Forest Products Association produces design specifications for wood construction, joists and rafters, and the all-weather wood foundation system. NFPA cooperated with others in the production of test standards for formaldehyde emissions from plywood and particleboard.

National Institute of Building Sciences (NIBS)
1015 15th Street, NW
Suite 700
Washington, DC 20005
(202) 347-5710

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National Institute of Standards and Technology (NIST)
(Formerly National Bureau of Standards [NBS])
Center for Building Technology
Gaithersburg, MD 20899
(301) 975-2000

NIST is a primary research and coordination arm for standards and measurement work. The Institute develops voluntary product standards for softwood lumber, and construction and industrial plywood. For CPSC, NIST also maintains a standard for surface flammability of carpets and rugs. The Institute also administers the National Voluntary Laboratory Accreditation Program (NVLAP) and is the main U.S. contact for ILAC, the International Conference for Laboratory Accreditation, an international effort aimed at reciprocity of testing laboratory recognition. The National Center for Standards and Certification Information is a repository and inquiry point on U.S. and foreign standards with more than 250,000 standards in its collection.

National Particleboard Association (NPA)
18298 Premiere Court
Gaithersburg, MD 20879
(301) 670-0604

National Pest Control Association (NPCA)
8100 Oak Street
Dunn Loring, VA 22027
(703) 573-8330

National Roofing Contractors Association (NRCA)
One O'Hare Center, Suite 8030
6250 River Road
Rosemont, IL 60018
(312) 318-6722

Naval Facilities Engineering Command (NAVFAC)
Naval Publications and Printing Service
Detachment Office
700 Robbins Avenue
Philadelphia, PA 19111
(215) 697-3321

NAVFAC publishes design manuals, military handbooks, standard design drawings, definitive design drawings and other construction standards for U.S. Navy facilities.

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National Conference of States on Building Codes and Standards (NCS\BCS)
481 Carlisle Drive
Herndon, VA 22070
(703) 437-0100

National Sanitation Foundation (NSF)
P.O. Box 1468
Ann Arbor, MI 48105
313 769-8010

Northeastern Lumber Manufacturers Association (NLMA)
272 Tuttle Road
P.O. Box 87A
Cumberland Center, ME 04121
(207) 829-6901

Portland Cement Association (PCA)
5420 Old Orchard Road
Skokie, IL 60077
(312) 966-6200

Provides information on use of concrete in building and construction.

Prestressed Concrete Institute (PCI)
175 West Jackson Blvd
Suite 1859
Chicago, IL 60604
(312) 786-0300

PCI has developed standards for fire resistive design of prestressed concrete.

Scaffolding, Shoring and Forming Institute (SSFI)
(formerly Scaffolding & Shoring Institute)
1230 Keith Building
Cleveland, OH 44115
(216) 241-7333

Provides recommended safety requirements, criteria, and inspection procedures for scaffolding, shoring and forming work.

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Sheet Metal and Air-Conditioning Contractors National Association (SMACNA)
8224 Old Courthouse Road
Vienna, VA 22182
(703) 790-9890

Documents produced by SMACNA technical committees relate to a variety of duct work considerations, and are widely referenced by model building codes and Federal agencies. These standards are updated periodically and are sold directly.

Society of the Plastics Industry, Inc. (SPI)
355 Lexington Avenue
New York, NY 10017
(212) 370-7340

SPI is a trade and professional association for over 1200 companies in the plastics industry. SPI cooperates in the standards activities of other organizations such as ASTM, ANSI and NSF. It has also produced a standard for an enclosed room fire test.

Southern Building Code Congress, International (SBCCI)
900 Montclair Road
Birmingham, AL 35213-1206
(205) 591-1853
(800) 633-3876

SBCC is one of three model building code organizations in the U.S. and produces a complete set of model building codes that are regularly updated, and are adopted by many regulatory jurisdictions, primarily in the southern states. SBCC Standard codes include building, fire protection, plumbing, mechanical, gas, and housing.

Southern Forest Products Association (SFPA)
P.O. Box 52468
New Orleans, LA 70152
(504) 443-4464

Southern Pine Inspection Bureau (SPIB)
4709 Scenic Highway
Pensacola, FL 32504
(904) 434-2611

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Steel Joist Institute (SJI)
1205 48th Avenue North, Suite A
Myrtle Beach, SC 29577
(803) 449-0487

The Steel Joist Institute is an association for producers of open-web steel joists. SJI standardization documents include load tables for various types of steel joists, and standard specifications for joist girders.

Tile Council of America (TCA)
Box 326
Princeton, NJ 08542
609 921-7050

Publishes standards for ceramic tile.

Truss Plate Institute (TPI)
583 D'Onofrio Drive, Suite 200
Madison, WI 53719
(608) 833-5900

TPI members are manufacturers of metal plates used in the production of wood tress assemblies. The organization has developed standards and design specifications for metal plate-connected wood trusses and parallel chord wood trusses.

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062
(312) 272-8800

UL has been in business since 1894 as an independent, not-for-profit testing organization and developer of safety standards. UL has published over 600 standards, of which over 300 are ANSI approved. Over 20 UL standards have been adopted by the Department of Defense. Some prominent UL building standards are for fire doors, roof deck construction, fire and smoke properties of cables used in air handling spaces, and an outline for investigation of insulated wall construction.

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United States Bureau of Mines (USBM)
Department of the Interior
2401 E Street, NW
Washington, DC 20240
(202) 343-1100

The USBM Ringelman Smoke Chart has been referenced in regard to exhaust emissions.

United States Department of Health and Human Services (HHS) Public Health Service
Division of Facilities Development
Parklawn Building, Room 11A19
5600 Fishers Lane
Rockville, MD 20857
(301) 443-6850

HHS standards for construction and equipment are used for hospitals and for medical facilities.

U.S. Postal Service (USPS)
Office of Design and Construction
475 L'Enfant Plaza West, SW
Washington, DC 20260
(202) 268-3873

Veterans Administration (VA)
Chief Architectural Specifications Division (088B31)
Office of Facilities
811 Vermont Avenue NW
Washington, DC 20420
(202) 393-4120

Water Pollution Control Federation (WPCF)
601 Wythe Street
Alexandria, VA 22314
(703) 684-2400

Western Fire Chiefs Association (WFCA)
530 South Workman Mill Road
Whittier, CA 90601
(213) 699-0541

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List of Third Party Suppliers of Standards Information

BRS **Bibliographic Retrieval Services**
1200 Route 7
Latham, NY 12110
(518) 783-1161

The database includes U.S. Military (DOD) standards, specifications, military sheet form standards (MS's), Air Force-Navy Aeronautical Standards (AN's), Qualified Product Lists (QPL's), Handbooks, U.S. Federal (GSA) standards, specifications, commercial item descriptions (CID's), QPL's, and U.S. standards developed by private sector organizations. Also includes the NCSCI database of industry standards, the national standards of major foreign countries, and the international standards of ISO and IEC.

CADIS **CAD Information Systems (CADIS), Inc.**
P.O. Box 5302
Englewood, CO 80155
(303) 799-1311

Computerized databases of construction industry products, standards, guide specifications, and related information with CAD-ready graphics.

CSI Construction Specifications Institute
601 Madison Street
Alexandria, VA 22314
(703) 684-0300

DEC **Document Engineering Company, Inc.**
15210 Stagg Street
Van Nuys, CA 91405
(213) 873-5566
(800)-MIL-SPEC

All military standards, specifications, and related documents.

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DIS **Dialog Information Services**
3460 Hillview Avenue
Palo Alto, CA 94304
(415) 858-3792
(800) 334-2564

The database includes U.S. Military standards and specifications, Qualified Product Lists (QPL's), Handbooks, U.S. Federal (GSA) standards, specifications, commercial item descriptions (CID's), QPL's, and U.S. standards developed by private sector organizations.

Global **Global Engineering Documents,**
Division of Information Handling Services
280 McGaw Avenue
Irvine, CA 92714
(714) 261-1455
(800) 854-7179

All military and Federal standards, specifications, handbooks, Commercial Item Descriptors, Qualified Products Lists, directives and regulations. Also NATO standards and publications of NASA, NBS, NRC, COM, DOT, FAA, FCC, GPO, GSA, HUD, OSHA, and NTIS. Also many voluntary standards.

IHS **Information Handling Services**
15 Inverness Way East
Englewood, CO 80150
(303) 790-0600
(800) 525-7052

All military Standards, specifications and related documents, Federal standards, Federal Construction Regulations and Standards, FAA, VA, NRC. Also many voluntary standards including those of the American Concrete Association, American Gas Association, American Institute of Architects, ANSI, ASCE, ASHRAE, ASME, American Society for Quality Control, ASTM, American Welding Society, IES, NFPA, Pipe Fabrication Institute, SAE, UL., and ISO.

The Information Handling Service Industry and International Standards databases, and the National Institute of Standards and Technology (NIST) Voluntary Engineering Standards databases are available on-line from BRS and Dialog, as is the Military and Federal Specifications and Standards database. The approximate cost is $65 per connect hour.

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NCSCI **National Center for Standards and Certificates Information**
National Institute of Standards and Technology
Room A-629
Gaithersburg, MD 20899
(301) 975-4040

Provides information on domestic and foreign standards, regulations, certification, and other standards-related activities. Maintains a microform and hard copy collection of more than 250,000 engineering standards and specifications, regulations, certification rules, directories, reference books and special publications. Responds to inquiries on the existence, source, and availability of standards and related documents. Develops and publishes lists, bibliographies and indexes of standards and related information. Provides on-line search and retrieval of standards documents information. NCSCI does not provide copies of codes and standards but their collection is open to the public for research.

NSA **National Standards Association, Inc.**
1200 Quince Orchard Boulevard
Gaithersburg, MD 20878
(800) 638-8094

All military and Federal standards, specifications, and related documents listed in the DOD Index of Specifications and Standards. Also standards of ARI, ACI, ASME, ASTM, NEMA, NFPA, SAE, UL, ISO and others. Also British, German, and Japanese standards. The National Standards Association electronic standards database is available on-line through DIALOG.

NTIS **National Technical Information Service**
U.S. Department of Commerce
5285 Port Royal Road
Springfield, VA 22161
(703) 487-4636

Some military standards; also ANSI publications cited in government standards.

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NPFC **Naval Publications and Forms Center**
5801 Tabor Avenue
Philadelphia, PA 19120
(215) 697-5637

All military and Federal standards, specifications and related documents listed in the Department of Defense Index of Specifications and Standards.

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